

(25)

MANDATORY APPENDIX 2

RULES FOR BOLTED FLANGE CONNECTIONS WITH RING TYPE GASKETS

2-1 GENERAL

(a) The rules in this Appendix cover the minimum requirements for design of bolted flange connections with gaskets that are entirely within the circle enclosed by the bolt holes and with no contact outside this circle and are to be used in conjunction with the applicable requirements in Subsections A, B, C, and D of this Division.

(1) The hub thickness of weld neck flanges and the design of components in the bolted flange connection, other than the flange, shall be in accordance with the applicable rules of Subsection A and the Mandatory Appendices.

(2) These rules are not to be used for the determination of the thickness of tubesheets integral with a bolting flange as illustrated in Figure UW-13.2, sketches (h) through (l) or Figure UW-13.3, sketch (c).

(3) Nonmandatory Appendix S provides discussion on design considerations for bolted flange connections.

(4) Design requirements for bolted flange connections in 2-4 and Section VIII, Division 2, Part 4.16 shall be used in lieu of those previously listed in this Appendix. Tables 2-1-1 and 2-1-2 list the new locations for requirements formerly located in this Division.

(5) The Section VIII, Division 1 design requirements listed in Table 2-1-3 shall be used in lieu of the corresponding Section VIII, Division 2 requirements referenced in Part 4.16.

(6) When this Appendix is applied for design, the applicable section of Section VIII, Division 2, Part 4.16 shall be used in accordance with UG-16(a) and Mandatory Appendix 46.

(7) For vessels in lethal service or when specified by the user or the user's designated agent, the maximum bolt spacing, B_{smax} , and the bolt spacing correction factor, B_{sc} , shall be applied in calculating the flange moment for internal pressure.

(8) Successful service experience may be used as an alternative to the flange rigidity rules for fluid services that are nonlethal and nonflammable and designed within the temperature range of -20°F to 366°F (-29°C to 186°C) without exceeding design pressures of 150 psi (1 035 kPa).

(b) It is recommended that bolted flange connections conforming to the standards listed in UG-44(a) be used for connections to external piping. These standards may be used for other bolted flange connections and dished

covers within the limits of size in the standards and the pressure-temperature ratings permitted in UG-44(a). The ratings in these standards are based on the hub dimensions given or on the minimum specified thickness of flanged fittings of integral construction. Flanges fabricated from rings may be used in place of the hub flanges in these standards provided that their strength, calculated by the rules in this Appendix, is not less than that calculated for the corresponding size of hub flange.

(c) Except as otherwise provided in (b), bolted flange connections for unfired pressure vessels shall satisfy the requirements in this Appendix.

(d) The rules of this Appendix should not be construed to prohibit the use of other types of flanged connections, provided they are designed in accordance with good engineering practice and the method of design is acceptable to the Inspector. Some examples of flanged connections that might fall in this category are as follows:

(1) flanged covers as shown in Mandatory Appendix 1, Figure 1-6

(2) bolted flanges using full-face gaskets

(3) flanges using means other than bolting to restrain the flange assembly against pressure and other applied loads

2-2 MATERIALS

(a) Materials used in the construction of bolted flange connections shall comply with the requirements given in UG-4 through UG-14.

(b) Flanges made from ferritic steel and designed in accordance with this Appendix shall be full-annealed, normalized, normalized and tempered, or quenched and tempered when the thickness of the flange, t (see Section VIII, Division 2, Figure 4.16.1), exceeds 3 in. (75 mm).

(c) Material on which welding is to be performed shall be proved of good weldable quality. Satisfactory qualification of the welding procedure under Section IX is considered as proof. Welding shall not be performed on steel that has a carbon content greater than 0.35%. All welding on flange connections shall comply with the requirements for postweld heat treatment given in this Division.

(d) Hubbed flanges shall not be machined from plate or bar material [except as permitted in UG-14(c)] unless the material has been formed into a ring and the following additional conditions are met:

Table 2-1-1
Paragraph Cross-Reference List

Division 1 Paragraph	Division 2 Paragraph
2-3, Notation — deleted	4.16.13
2-4, Circular Flange Types	4.16.3
2-5, Bolts Loads — deleted	4.16.2 and 4.16.6
2-6, Flange Moment — deleted	4.16.7
2-7, Calculation of Flange Stresses — deleted	4.16.7
2-8, Allowable Flange Design Stress — deleted	4.16.7
2-9, Split Loose Flanges — deleted	4.16.8
2-10, Noncircular Shaped Flanges with Circular Bore — deleted	4.16.9
2-11, Flanges Subject to External Pressure — deleted	4.16.7
2-12, Flanges With Nut Stops — deleted	4.16.10
2-13, Reverse Flanges — deleted	4.16.3.2 and 4.16.7
2-14, Flange Rigidity — deleted	4.16.7

(1) In a ring formed from plate, the original plate surfaces are parallel to the axis of the finished flange. (This is not intended to imply that the original plate surface should be present in the finished flange.)

(2) The joints in the ring are welded butt joints that conform to the requirements of this Division. Thickness to be used to determine postweld heat treatment and radiography requirements shall be the lesser of

$$t \text{ or } \frac{(A - B)}{2}$$

where these symbols are as defined in 2-3.

(3) The back of the flange and the outer surface of the hub are examined by either the magnetic particle method as per [Mandatory Appendix 6](#) or the liquid penetrant method as per [Mandatory Appendix 8](#).

(e) Bolts, studs, nuts, and washers shall comply with the requirements in this Division. It is recommended that bolts and studs have a nominal diameter of not less than $\frac{1}{2}$ in. (13 mm). If bolts or studs smaller than $\frac{1}{2}$ in. (13 mm) are used, ferrous bolting material shall be of alloy steel. Precautions shall be taken to avoid over-stressing small-diameter bolts.

2-4 CIRCULAR FLANGE TYPES

(a) *Integral-Type Flanges.* Integral-type flanges are addressed in Section VIII, Division 2, 4.16.3.1(a).

(b) *Welded Slip-on-Type Flanges.* Welded slip-on-type flanges are addressed in Section VIII, Division 2, 4.16.3.1(b).

(c) *Loose-Type Flanges.* Loose-type flanges are addressed in Section VIII, Division 2, 4.16.3.1(c).

(d) Reverse-Type Flanges

(1) *Integral-Type Reverse Flange.* The shell-to-flange attachment of integral-type reverse flanges may be attached as shown in Section VIII, Division 2, Table 4.2.9, details (6) through (10), or Figure 2-4, as well as Figure UW-13.2, sketches (a) and (b). The requirements of 2-4(c) apply to Figure 2-4, as well as Figure UW-13.2, sketches (a) and (b).

(2) *Welded Slip-on-Type Reverse Flange.* The shell-to-flange attachment of welded slip-on-type reverse flanges may be attached as shown in Section VIII, Division 2, Table 4.2.9, details (2) and (4), or Figure 2-4, as well as Figure UW-13.2, sketches (c) and (d). When Figure UW-13.2, sketches (c) and (d) are used, the maximum wall thickness of the shell shall not exceed $\frac{3}{8}$ in. (10 mm), and the maximum design metal temperature shall not exceed 650°F (340°C).

2-15 QUALIFICATION OF ASSEMBLY PROCEDURES AND ASSEMBLERS

It is recommended that flange joints designed to this Appendix be assembled by qualified procedures and by qualified assemblers. ASME PCC-1 may be used as a guide.

Table 2-1-2
Figure and Table Cross-Reference List

Topic	Division 1	Division 2
Loose-type flange	Figure 2-4, sketch (1) — deleted	Figure 4.16.5
	Figure 2-4, sketch (1a) — deleted	Table 4.2.9, detail 5
	Figure 2-4, sketches (2) and (2a) — deleted	Screwed Type Flanges — deleted
	Figure 2-4, sketch (3) — deleted	Figure 4.16.2; Table 4.2.9, detail 1 (welded slip-on flange)
	Figure 2-4, sketch (3a) — deleted	Figure 4.16.1(b); Table 4.2.9, detail 2 (integral flange)
	Figure 2-4, sketch (4) — deleted	Figure 4.16.2; Table 4.2.9, detail 3 (welded slip-on flange)
	Figure 2-4, sketch (4a) — deleted	Figure 4.16.1(b); Table 4.2.9, detail 4 (integral flange)
	Figure 2-4, sketch (4b) — deleted	Figure 4.16.1(b); Table 4.2.9, detail 11 (integral flange)
	Figure 2-4, sketch (4c) — deleted	Figure 4.16.2; Table 4.2.9, detail 12 (welded slip-on flange)
Integral-type flange	Figure 2-4, sketch (5) — deleted	Figure 4.16.1(c)
	Figure 2-4, sketch (6) — deleted	Figure 4.16.1(d); Table 4.2.9, detail 6
	Figure 2-4, sketch (6a) — deleted	Figure 4.16.1(e); Table 4.2.9, detail 7
	Figure 2-4, sketch (6b) — deleted	Figure 4.16.1(f); Table 4.2.9, detail 8
	Figure 2-4, sketch (7) — deleted	Figure 4.16.1(a); Table 4.2.9, detail 9
Optional type flange	Figure 2-4, sketches (8), (8A), (9), (9a), (10), (10a), and (11) — deleted	Optional type flange — deleted
Flange with nut stop	Figure 2-4, sketch (12)	Figure 4.16.3
	Figure 2-4, sketch (12a)	Figure 4.16.4
Minimum contact width	Table 2-4	Table 4.16.2
Gasket factors	Table 2-5.1	Table 4.16.1
Effective gasket width	Table 2-5.2	Table 4.16.3
Location of gasket reaction	Table 2-5.2	Figure 4.16.8
Moment arms for flange loads	Table 2-6	Table 4-16.6
Terms involving K	Figure 2-7.1	Table 4.16.4
Flange factors in formula form	Table 2-7.1	Table 4.16.5
Values of F for integral flange	Figure 2-7.2 — deleted	Table 4.16.5
Values of V for integral flange	Figure 2-7.3 — deleted	Table 4.16.5
Values of F_L for loose flange	Figure 2-7.4 — deleted	Table 4.16.5
Values of V_L for loose flange	Figure 2-7.5 — deleted	Table 4.16.5
Values of f	Figure 2-7.6 — deleted	Table 4.16.5
Integral-type reverse flanges	Figure 2-13.1	Figure 4.16.6(a)
Loose-ring reverse flange	Figure 2-13.2	Figure 4.16.7
Flange rigidity	Table 2-14	Table 4.16.10

Table 2-1-3
Division 2 Cross-Reference List to Division 1 Requirements

Topic	Division 2		Division 1
	Paragraph	Reference	
Flange and pipe fittings	4.16.1.3	4.1.11	UG-44(a)
Design rules for flange joints	4.16.1.4	Part 5	2-1(d), U-2(g)
Flange materials	4.16.4	Part 3	2-2
Joint assembly procedure	4.16.11	...	2-15
Evaluation of external forces and moments for flanged joints with standard flanges	4.16.12	4.1.11	UG-44(b)

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